

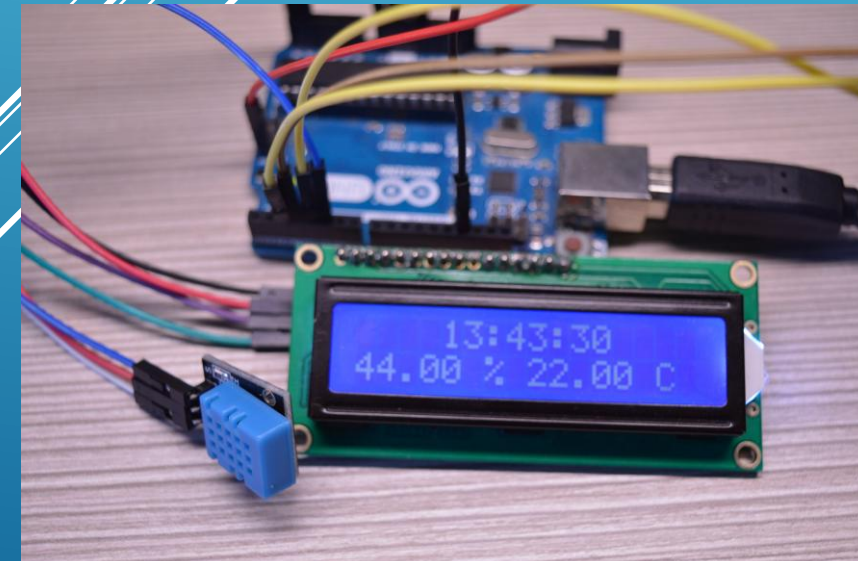
Arduino Weather Station

ECS3001

Project Guide: Prof. Ramesh Prasad Panda

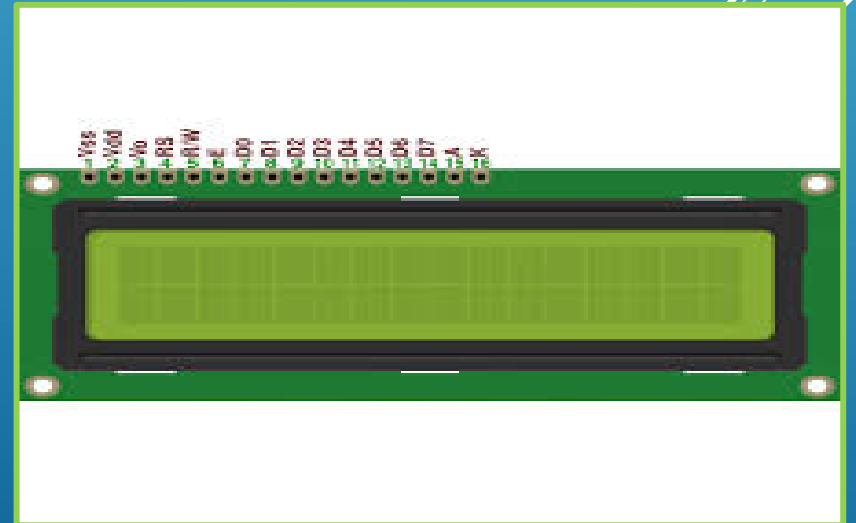
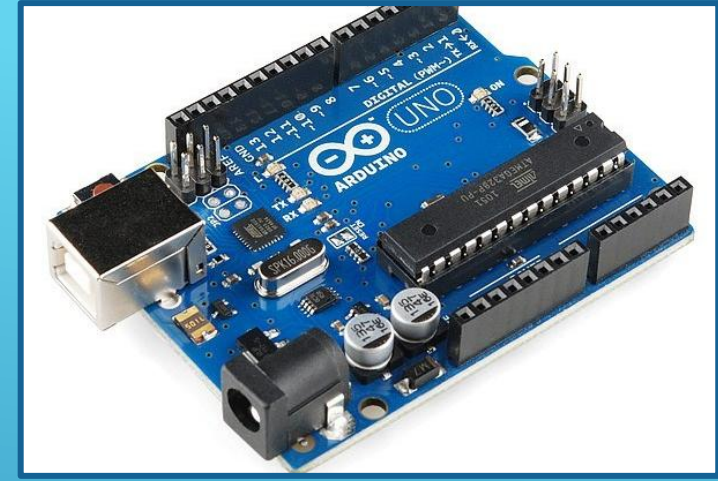
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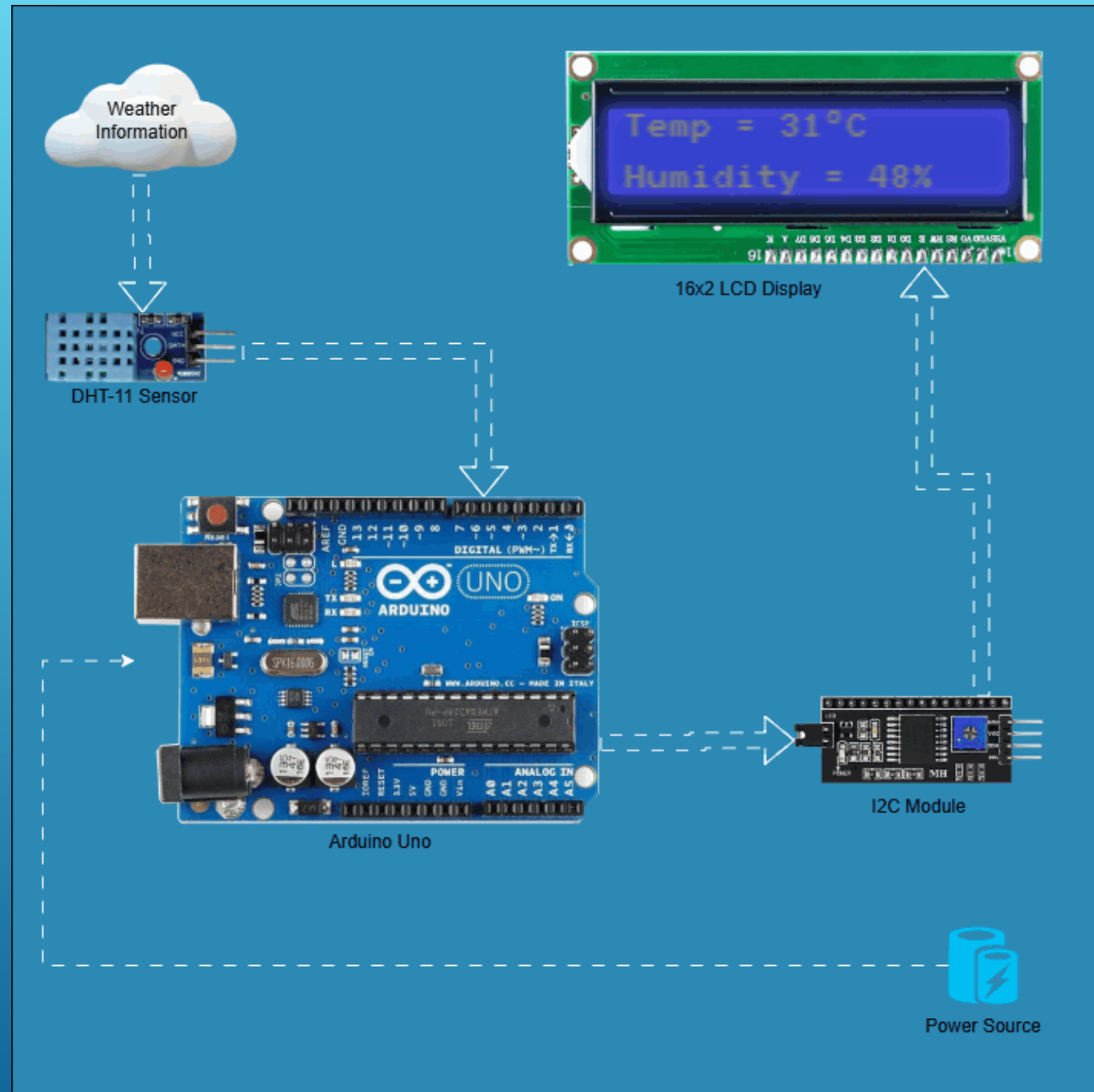
Components Used:

- **Arduino Uno:** The main control unit of the weather station. It reads data from various environmental sensors, processes the information, and controls the output devices such as the display.
- **16×2 LCD Display:** presents real-time weather data such as temperature, humidity, and other parameters in a clear and readable format.
- **I2C module:** interfaces with the 16×2 LCD display with the Arduino Uno using only the two communication lines (SDA and SCL).
- **DHT11 Sensor:** measures the temperature and humidity, providing digital output data, making it easy to interface with the Arduino Uno.



Function:

- Whenever the system is powered on, the DHT11 sensor begins after a short delay detecting the surrounding temperature and humidity of the environment.
- The Arduino UNO then reads the data from the DHT11 sensor and sets the control for the display system of the weather station.
- The processed temperature and humidity values are transmitted to the 16×2 LCD display through the I2C module, set to appear in regions where values may dynamically be updated over time.
- The LCD screen continuously shows real-time temperature and humidity readings, allowing users to monitor environmental conditions easily for as long as the power is being supplied to the system.



Circuit Setup:

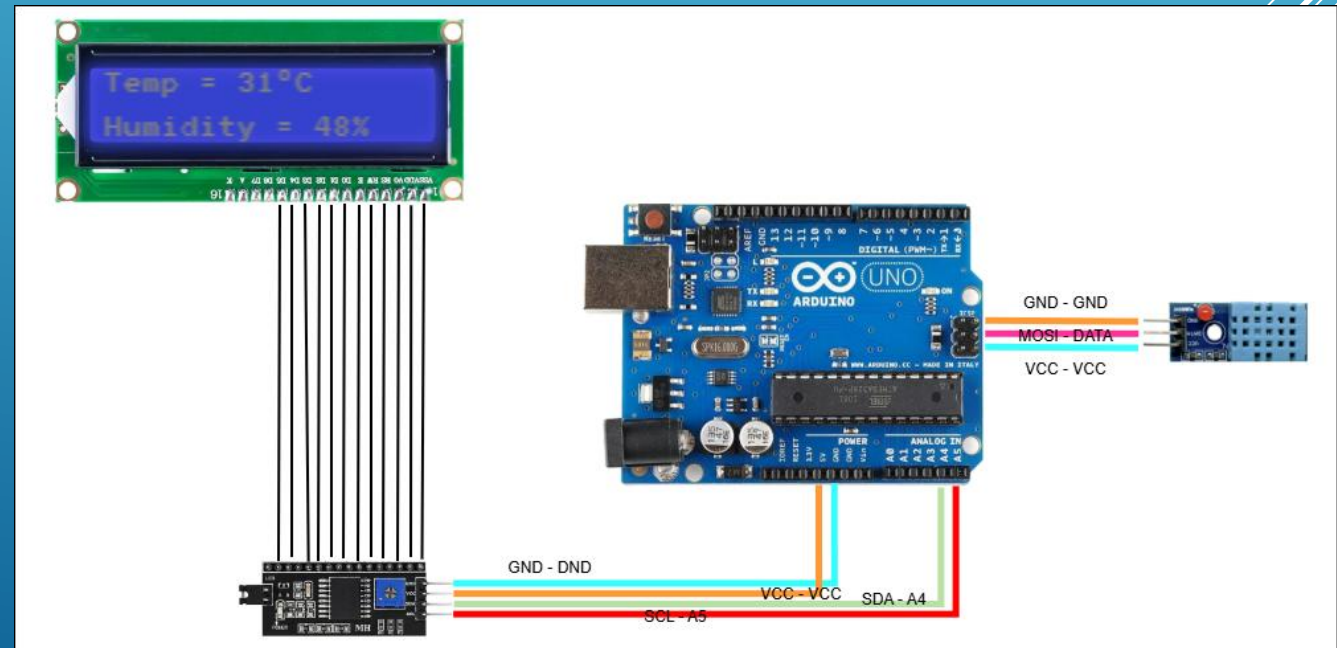
With the Arduino Uno acting as the central controller, all electronic components are connected as follows:

I2C Module:

- I2C backpack GND → Arduino GND: common ground reference.
- I2C backpack VCC → Arduino 5V: powers the I2C LCD interface.
- I2C backpack SDA → Arduino A4: carries I2C data between Arduino and LCD.
- I2C backpack SCL → Arduino A5: carries I2C clock signal for synchronization.

DHT-11 Sensor:

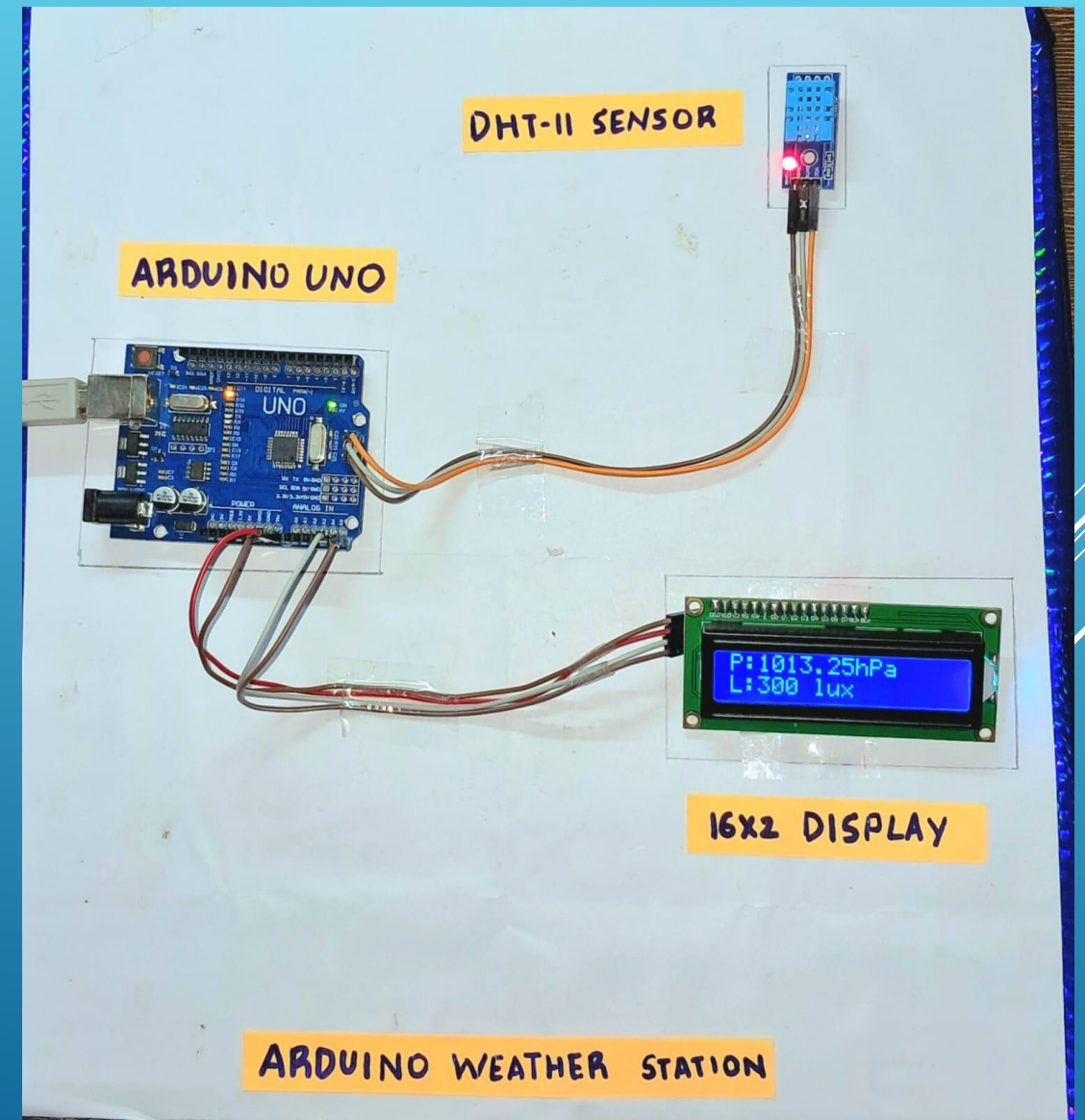
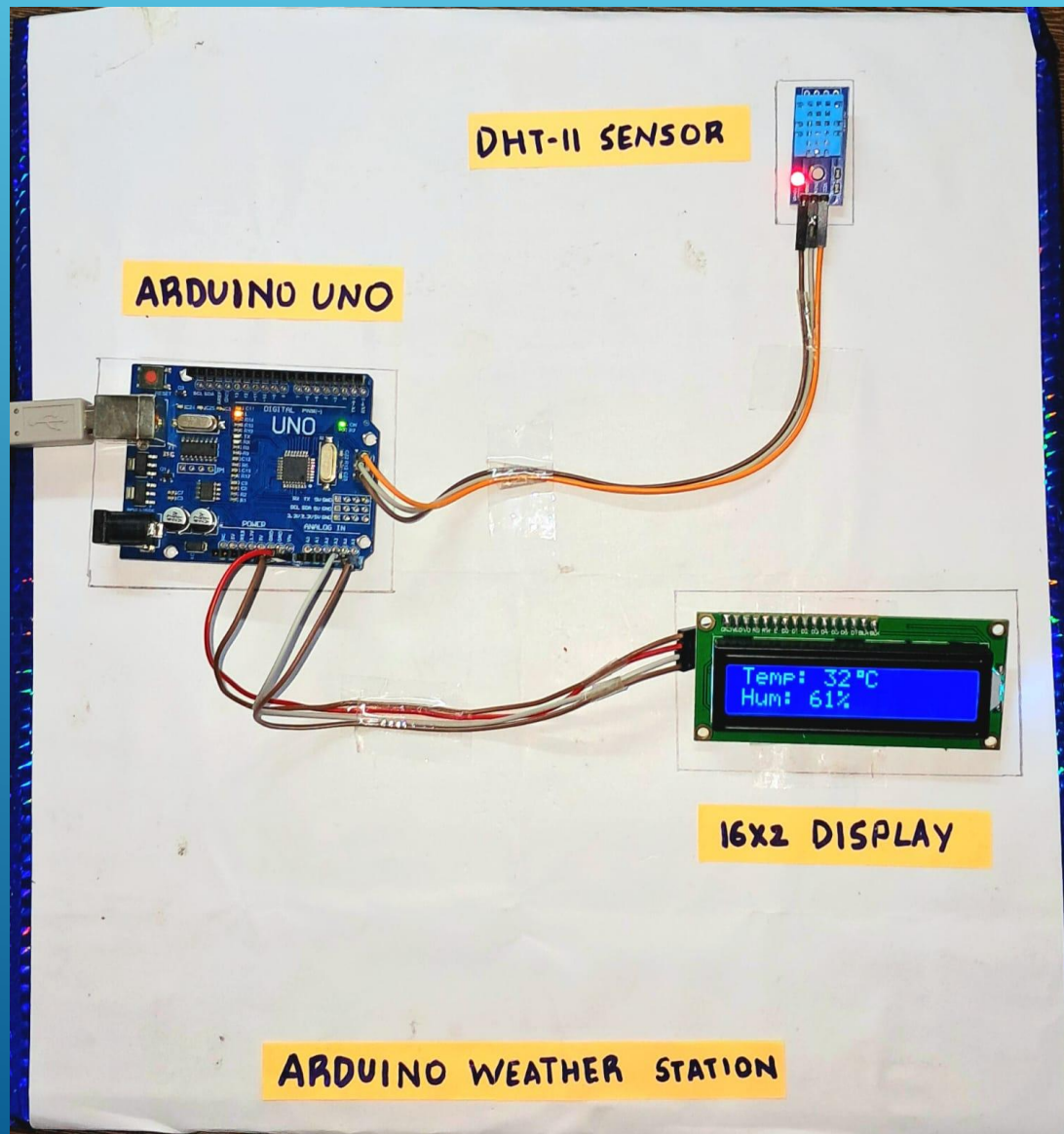
- Temperature/Humidity sensor VCC → Arduino 5V: supplies power to the sensor.
- Temperature/Humidity sensor GND → Arduino GND: completes power circuit for the sensor.
- Temperature/Humidity sensor DATA → Arduino digital pin (e.g., D2): sends measured temperature and humidity data to the Arduino.



Results:

- The prototype of the Arduino Weather Station has been successfully assembled and tested under controlled indoor/outdoor conditions.
- The DHT-11 Sensor measures both temperature and humidity in one device, so we don't need separate sensors, making it easy to interface with an Arduino Uno.
- The use of the I2C module reduces wiring drastically (only 4 wires: VCC, GND, SDA, SCL instead of many parallel pins) and allows us to use multiple devices on the same two wires.
- The present implementation involves using an external power supply (typically 7–12V), so the circuit can run without a computer, enabling portable or standalone operation (e.g., wall adapter or battery pack).
- The Arduino-based control system and communication module works reliably, allowing real-time monitoring and minor adjustments during operation.
- Overall, the prototype demonstrates energy-efficient, low-maintenance, and cost-efficient weather monitoring.

Final Prototype:



Conclusion:

- The Arduino-based control system ensures reliable autonomous operation, with easy adaptability and scalability for future upgrades.
- Test results confirm that the system can autonomously monitor multiple environmental parameters, including temperature, humidity, light intensity, and atmospheric pressure, and display these readings efficiently, validating the project's objectives of functionality and performance.
- Overall, the project lays a strong foundation for future research in advanced weather monitoring systems, promoting practical and environmentally conscious solutions for real-world applications such as agriculture, smart environments, surveillance, and disaster monitoring.

THANK YOU

